Khulna University of Engineering & Technology

Department of Electronics and Communication Engineering

Course No: ECE 2200

Course Title: Electronic Circuits Design Laboratory

Design an Adjustable, Regulated, and Protected DC Power Supply

Group No: B-6

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1. Introduction

A regulated DC power supply is an essential device in electronics laboratories, research institutions, industries, and educational environments. Most electronic circuits require a stable and adjustable DC voltage for proper operation. However, fluctuations in the input supply, excessive output voltage, low output voltage, and accidental short circuits may damage sensitive electronic components and reduce system reliability.

This project presents the design and implementation of an adjustable regulated DC power supply incorporating **Over Voltage Protection (OVP), Under Voltage Protection (UVP), and Short Circuit Protection (SCP)**. The system converts the 220 V AC mains supply into a regulated low-voltage DC output through a step-down transformer, bridge rectifier, filter capacitor, and LM317 adjustable voltage regulator. Protection circuits based on the LM393 voltage comparator continuously monitor the output voltage and isolate the load using a relay whenever abnormal operating conditions occur.

The proposed power supply offers adjustable output voltage, improved voltage regulation, enhanced operational safety, and reliable protection against electrical faults. It is suitable for powering laboratory experiments, embedded systems, electronic prototypes, and testing various low-voltage electronic devices.

2. Objectives

The primary objectives of this project are:

i. To design and implement an adjustable regulated DC power supply using the LM317 voltage regulator.
ii. To convert 220 V AC mains voltage into a smooth and regulated DC output.
iii. To provide an adjustable output voltage suitable for different electronic applications.
iv. To implement Over Voltage Protection (OVP) to protect connected devices from excessive voltage.
v. To implement Under Voltage Protection (UVP) for maintaining reliable operation.
vi. To implement Short Circuit Protection (SCP) to prevent excessive current flow during fault conditions.
vii. To develop a relay-based isolation mechanism that disconnects the load whenever abnormal conditions are detected.
viii. To improve the safety, efficiency, and reliability of conventional laboratory power supplies.
ix. To evaluate the performance of the designed system under various operating and fault conditions.

3. Theory

The proposed power supply consists of several functional stages that operate together to produce a stable and protected DC output.

AC to DC Conversion

The system receives a **220 V AC** input from the mains supply. A **24 V, 2 A** step-down transformer reduces the voltage to a safe level suitable for electronic circuits.

The reduced AC voltage is converted into pulsating DC using a full-wave bridge rectifier consisting of four **1N4007** diodes. During each half-cycle of the AC waveform, two diodes conduct, allowing current to flow in the same direction through the load.

Filtering

The rectified output contains ripple components. A **4700 µF** electrolytic capacitor smooths the pulsating DC by charging during voltage peaks and discharging during voltage drops, thereby reducing ripple voltage.

Voltage Regulation

An **LM317** adjustable voltage regulator is used to obtain a stable DC output. The output voltage can be adjusted by varying the resistor network according to

$$V_{out}=1.25(1+R_2/R_1)+I_{adj}R_2$$

where

- R1 is typically 240 Ω,
- R2 is the adjustable resistor,
- Iadj is very small and usually neglected.

Over Voltage Protection (OVP)

An LM393 comparator continuously compares the output voltage with a preset reference voltage. If the output exceeds the safe limit, the comparator activates a transistor that de-energizes the relay, disconnecting the load from the power supply.

Under Voltage Protection (UVP)

Another comparator monitors the minimum allowable output voltage. If the voltage falls below the preset threshold, the relay disconnects the load, preventing improper operation of sensitive electronic devices.

Short Circuit Protection (SCP)

During a short circuit, the output voltage drops rapidly while current tends to increase significantly. The protection circuit detects this abnormal condition and immediately disconnects the load using the relay, thereby preventing overheating and damage to the regulator and connected components.

Relay Protection

A transistor-based relay driver energizes the 12 V relay coil during normal operation. A flyback diode connected across the relay protects the transistor from the reverse voltage generated when the relay is switched off.

4. Motivation & Background

Laboratory experiments and electronic circuit development require a reliable DC power source capable of supplying different voltage levels without introducing excessive ripple or voltage fluctuations. Although many commercial power supplies are available, they often lack comprehensive protection features or are relatively expensive for educational laboratories.

Traditional power supplies mainly focus on voltage regulation while providing limited protection against abnormal operating conditions. Overvoltage may permanently damage sensitive integrated circuits, undervoltage can lead to unstable circuit behavior, and short circuits may cause excessive current flow that damages components and power devices.

The motivation behind this project is to develop a cost-effective, adjustable, and protected DC power supply that combines voltage regulation with multiple protection mechanisms. The project also provides practical experience in analog electronics, voltage regulation, comparator circuits, transistor switching, relay control, and power electronics. The resulting system offers improved safety, reliability, and versatility for laboratory experiments and electronic prototyping.

5. Required Apparatus

Table: The following components and instruments are required to implement the project:

SL No.	Component	Specification	Purpose	Quantity
1	Step-down Transformer	220 V AC / 24 V AC, 2 A	Voltage conversion	1 pc
2	1N4007 Diodes	1000 V, 1 A	Bridge Rectifier (AC to DC conversion)	4 pcs
3	Filter Capacitors	4700 µF, 10 µF, 0.33 µF, 0.1 µF	Ripple reduction and regulator stability	As required
4	LM317T	Adjustable Voltage Regulator	Output voltage regulation	1 pc
5	LM393	Dual Voltage Comparator	OVP and UVP detection	1 pc
6	IC 7812	12 V Voltage Regulator	Comparator and relay supply	1 pc
7	Zener Diodes	6.2 V and 3.3 V	Reference voltage generation	2 pcs
8	Relay (SPDT)	12 V Coil	Load isolation	1 pc
9	Transistors	BD135, BD136, BC557, 2N2222A	Relay driver and switching	As required

10	Potentiometers	10 kΩ, 50 kΩ	Voltage adjustment and threshold setting	3 pcs
11	Resistors	Various values	Biasing and voltage division	Assorted
12	LEDs	Red and Green	Status indication	2 pcs
13	Breadboard	—	Circuit assembly	2 pc
14	Digital Multimeter	0~600V, 10A	Voltage measurement	1 pc
15	Oscilloscope	—	Waveform observation	1 pc

6. Circuit Diagram & Analysis (Experimental Setup)

6.1 Circuit Description

The proposed circuit consists of four functional sections:

1. **Power Conversion Stage**
    - A 220 V/24 V transformer reduces the mains voltage.
    - A bridge rectifier converts AC into pulsating DC.
    - A 4700 μF capacitor filters the output to produce smooth DC.
2. **Voltage Regulation Stage**
    - The LM317 regulator provides an adjustable output voltage.
    - Output voltage is controlled using a resistor network and potentiometer.
3. **Protection Stage**
    - LM393 comparators continuously monitor the output voltage.
    - Reference voltages are generated using Zener diodes and adjustable potentiometers.
    - When the output exceeds or falls below preset limits, the comparator output changes state.
4. **Relay Control Stage**
    - Comparator outputs drive transistor switches.
    - The transistor energizes or de-energizes a 12 V relay.
    - The relay disconnects the load during overvoltage, undervoltage, or short-circuit conditions.
    - LED indicators display normal operation and fault status.

The complete system ensures stable voltage regulation while providing automatic protection against electrical faults, making it suitable for powering sensitive electronic circuits.

6.2 Experimental Setup

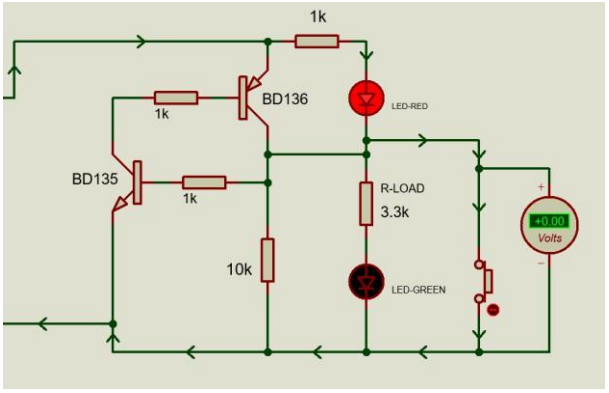


Fig1: short circuit protection

(when output is shorted)

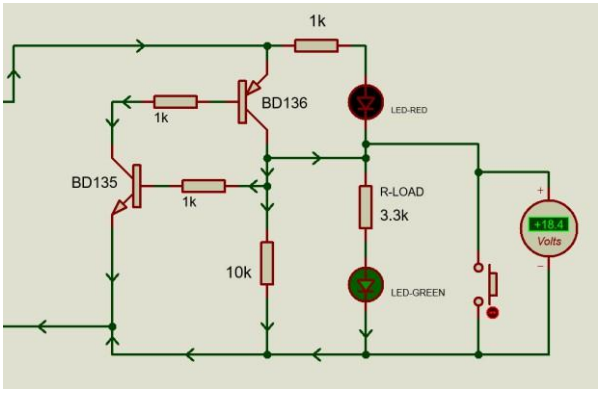


Fig2: short circuit protection

(when output is not shorted)

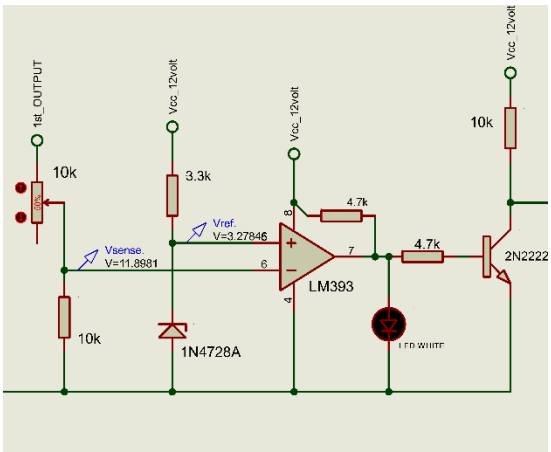


Fig3: Under-Voltage protection circuit

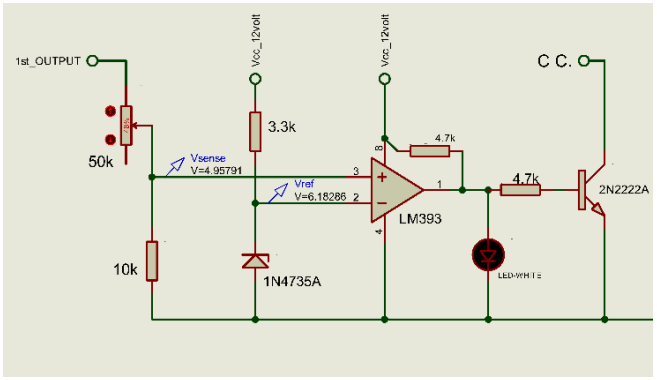


Fig4: Over-Voltage protection circuit

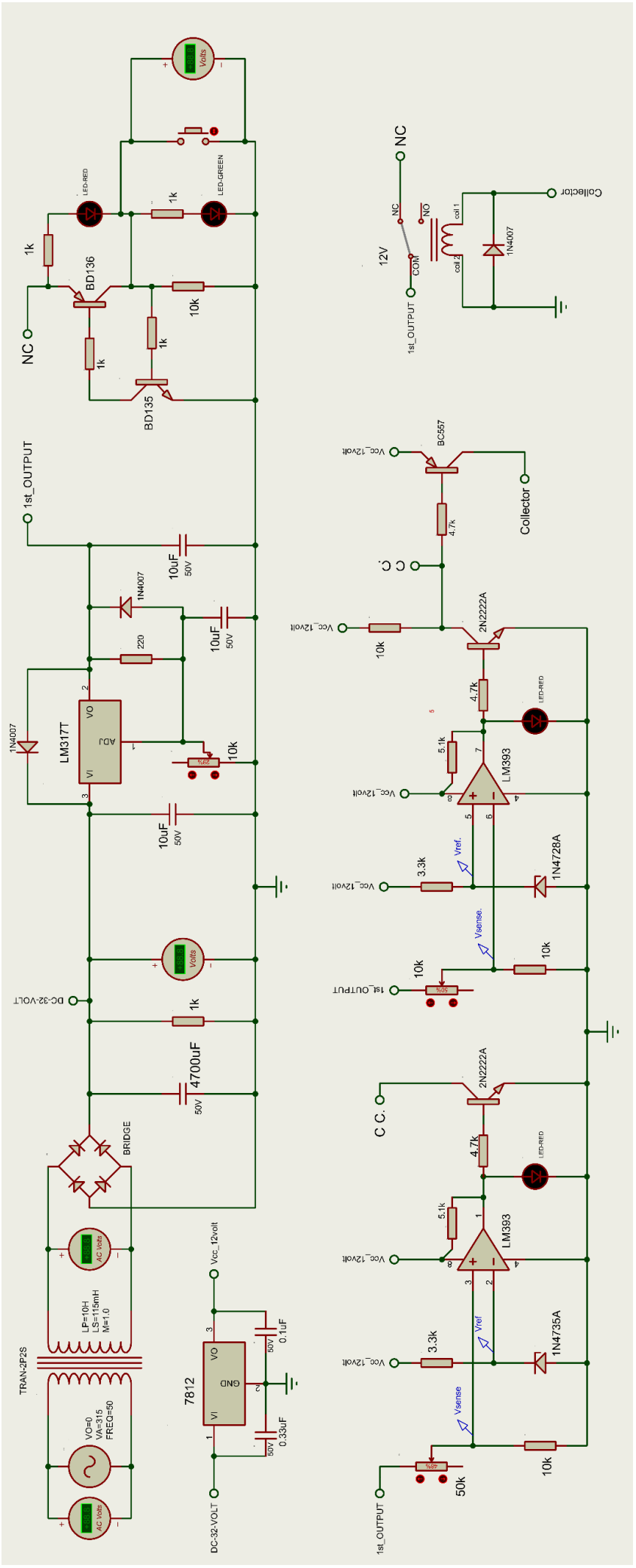


Fig5: circuit diagram for Adjustable Regulated DC Power Supply

7. Results & Discussion

The developed power supply successfully produced a stable and adjustable DC output voltage suitable for laboratory applications. The LM317 regulator maintained nearly constant output voltage despite moderate variations in input voltage and load conditions.

Experimental testing showed that:

- The bridge rectifier and filter capacitor effectively reduced ripple voltage.
- The LM317 regulator provided smooth and adjustable output voltage.
- The Overvoltage Protection (OVP) circuit disconnected the load whenever the output exceeded the preset threshold.
- The Undervoltage Protection (UVP) circuit isolated the load during low-voltage conditions.
- The Short-Circuit Protection (SCP) mechanism rapidly disconnected the load, preventing excessive current flow.
- LED indicators accurately displayed the operating status of the system.

Overall, the implemented system demonstrated good voltage regulation, reliable protection performance, and satisfactory operational stability.

8. Socio-Economic Impact and Sustainability

The proposed power supply offers several social, economic, and environmental benefits.

- Improves laboratory safety by preventing equipment damage.
- Reduces maintenance and replacement costs of electronic devices.
- Extends the service life of electronic equipment through reliable protection mechanisms.
- Provides an affordable laboratory power supply suitable for educational institutions.
- Encourages local manufacturing using readily available components.
- Consumes relatively low power and supports sustainable utilization of electronic resources.
- Reduces electronic waste by minimizing component failures caused by electrical faults.

9. Project Planning

The project was completed according to the following schedule:

1st Week	Study the concept of adjustable power supplies and protection systems (OVP, UVP, SCP), select components, and prepare the circuit design.
2nd Week	Design and simulate the complete circuit in Proteus, verify voltage regulation and protection operation under different fault conditions.
3rd Week	Assemble the hardware on breadboard/PCB, test individual modules (rectifier, regulator, OVP, UVP, SCP, relay circuit).
4th Week	Integrate the complete system, perform final testing, troubleshoot issues, collect results, and prepare the project report and presentation.

10. Budget Analysis

Table: table for the item's prices and total budget

SL No.	Item	Description	Cost (BDT)
1	Transformer	1 × 350/-	350
2	Capacitors	(1×55/-)+(3×5/-)+(1×5/-)+(1×2/-)	77
3	Resistors	18 × 0.7 /-	13
4	Potentiometers	3 × 25/-	75
5	Diodes	7 × 4/-	28
6	Zener diodes	2 × 5/-	10
7	IC 7812	1 × 15/-	15
8	LM393	1 × 15/-	15
9	Relay	1 × 38/-	38
10	LM317T	1 × 32/-	32
11	NPN transistors	(2 × 3/-)+(1 × 5/-)	11
12	PNP transistors	(1 × 5/-)+(1 × 5/-)	10
13	Voltmeter	1 × 70/-	70
14	Miscellaneous	-	50
Total: 794 TK **in words:** Seven hundred and ninety-four taka only			

The total estimated project cost is **794 BDT**, making the design economical and suitable for laboratory and educational applications. Most components are inexpensive and readily available in local electronics markets, reducing implementation costs while maintaining reliable performance.

11. Conclusion

This project successfully demonstrated the design and implementation of an adjustable regulated DC power supply with integrated Overvoltage Protection (OVP), Undervoltage Protection (UVP), and Short-Circuit Protection (SCP). The developed system provides a stable and adjustable DC output while ensuring the safety of connected electronic equipment through automatic fault detection and load isolation.

The experimental results verified that the power supply achieved satisfactory voltage regulation, effective protection performance, and reliable operation under various operating conditions. The use of the LM317 voltage regulator, LM393 comparators, transistor switching, and relay-based protection resulted in a simple, economical, and dependable design suitable for laboratory and educational applications.

Overall, the project successfully fulfilled its objectives and demonstrated the practical application of analog electronic principles in developing a safe, efficient, and low-cost regulated DC power supply.